CHAPTER 7

Recent trends and digital technology applications in lower limb injury rehabilitation

Saghil Sali[1], Rifai Chai[2] and Balasankar Ganesan[3]
[1]Institute of Health and Sports, Victoria University, Melbourne, VIC, Australia
[2]School of Science, Computing and Engineering Technologies (SoSCET), Swinburne University of Technology, Melbourne, VIC, Australia
[3]Discipline of Occupational Therapy, Institute of Health and Wellbeing, Federation University, Churchill (Gippsland Campus), VIC, Australia

Learning objectives

- Understanding the latest digital technology applications used in lower limb injury rehabilitation and their potential benefits for patients.
- Familiarizing oneself with virtual reality, wearable devices, mobile applications, and upcoming digital technologies used in lower limb injury rehabilitation.
- Gaining knowledge about how digital technology applications can be used to personalize rehabilitation programs, enhance patient engagement and applied recovery methodologies .
- Understanding the potential limitations and challenges of digital technology applications in lower limb injury rehabilitation.
- Gaining the ability to critically evaluate research studies related to digital technology applications in lower limb injury rehabilitation and further assess their validity and reliability.
- Learning about the importance of interdisciplinary collaboration in designing and implementing effective digital technology applications in lower limb rehabilitation.
- Understanding the ethical and legal considerations related to the use of digital technology applications in lower limb rehabilitation.

7.1 Introduction

The medical device industry has been experiencing a surge in innovation, driven by the need for homecare, preventative treatments, early diagnosis, reducing patient recovery times, and improving outcomes, as well as the emergence of technologies such as machine learning (ML), augmented reality (AR), 5G, and digitalization (GlobalData, 2021). According to GlobalData's report on Robotics in Medical Devices: Robotic Lower Limb Rehabilitation, over 450,000 patents have been filed and granted in the medical device industry in the last 3 years alone. However, not all innovations in the medical devices industry follow a constant upward trend. Rather,

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their evolution is characterized by an S-shaped curve that reflects their typical lifecycle, from early emergence to accelerating adoption, before finally stabilizing and reaching maturity (Luo, Li, & Lu, 2020). Therefore, identifying where a particular innovation stands on this journey, especially those in the emerging and accelerating stages, is critical for understanding their current level of adoption and the likely future trajectory and impact they will have on the industry.

Lower limb injuries can have a significant impact on an individual's mobility and quality of life. Traditionally, rehabilitation for lower limb injuries involved a combination of physiotherapy, occupational therapy, and exercises (Egan, Maher, & Sherrington, 2020). However, recent advancements in digital technology have opened new avenues for rehabilitating lower limb injuries. Digital technology applications such as virtual reality (VR), wearable devices, and mobile applications are being increasingly used in the rehabilitation process, providing new ways to monitor patient progress, personalize rehabilitation programs, and enhance patient engagement (Bashshur, Shannon, & Krupinski, 2016). These technologies have the potential to improve the effectiveness and efficiency of rehabilitation programs, leading to faster recovery times and better outcomes for patients. In this essay, we will explore recent trends and digital technology applications in lower limb injuries rehabilitation and examine their potential benefits and limitations.

7.2 Immersive technology

Immersive technologies, such as VR/AR, are rapidly expanding in the healthcare industry, providing new opportunities to make therapies more engaging and effective for patients (VentureBeat, 2021). Startups are providing tailored software and hardware tools for VR/AR therapies, allowing healthcare professionals to create personalized treatment plans for patients. For instance, AR/VR therapies have been developed, such as climbing games for people undergoing upper-limb rehabilitation (Kopelman, Selles, & Janssen, 2021). Additionally, virtual coaching within a virtual space has shown promising results in allowing patients to immerse themselves in therapy sessions more easily (Ustinova, Perkins, & Leonard, 2020). The use of immersive technology in rehabilitation has broken down barriers to entry for patients, making it more accessible and convenient than traditional rehabilitation methods (VentureBeat, 2021). Immersive technology has become the most prominent trend in rehabilitation technologies, providing benefits such as increased engagement, motivation, and personalization (Ustinova et al., 2020). Immersive technologies such as VR/AR provide a new opportunity for healthcare professionals to create personalized treatment plans for patients. The potential benefits of immersive technology in rehabilitation include increased engagement, motivation, and convenience. This technology has the potential to revolutionize the rehabilitation industry and improve patient outcomes. Villiger et al. (2017) conducted a study on the effectiveness of VR training in improving lower limb muscle strength, balance, and functional mobility in individuals with chronic incomplete spinal cord injury. The VR training setup simulated involved various scenarios for training different lower limb muscles and functions. For example, ankle dorsal flexion was trained through scenarios such as Hamster Splash, which involved launching hamsters into a swimming pool (A); Footbag, which involved juggling a ball (B); and Get to the Game, which involved walking from home via a tram station to a stadium (E). Knee extension was trained through scenarios such as Star Kick, which involved kicking balls toward stars (C), while leg ad-/abduction was trained through scenarios such as Planet Drive, which involved avoiding touching oncoming cars (D). The study found that VR-augmented training led to significant improvements in upper and lower limb muscle strength, balance, and functional mobility in individuals with chronic incomplete spinal cord injury.

7.2.1 Designs for rehabilitation environments by dynamics virtual reality

Dynamics VR, a Spanish startup, is making physiotherapy more accessible by developing specialized environments for both rehabilitation patients and professionals (European Startup Prize, 2020). The company's software solution is designed to be used with commercially available VR sets, providing clients with an affordable, easy-to-use, and comprehensive technology for successful rehabilitation.

The startup's solution aims to reduce kinesiophobia and pain by 37% and 60%, respectively (Goudman, 2022; Moon, Huang, Sadeghi, Akinwuntan, & Devos, 2021; European Startup Prize, 2020). By creating a safe and immersive environment, patients can undergo rehabilitation sessions in a comfortable and engaging manner, increasing the likelihood of successful outcomes. Additionally, the solution extends the duration of rehabilitation sessions, making them more effective in the long run (Goudman, 2022). Using VR technology, Dynamics VR is providing a unique solution for rehabilitation that addresses the challenges of traditional physiotherapy. The startup's approach is innovative, affordable, and convenient, enabling more individuals to benefit from rehabilitation services. Moon et al. (2021) presented the proof of concept of the Virtual Reality Comprehensive Balance Assessment and Training (VR-ComBAT). The setup includes a VR headset and a foam surface placed over a force plate for a participant to stand on. The VR-ComBAT aims to provide a comprehensive balance assessment and training for the sensory organization of dynamic postural control. This setup enables participants to experience different sensory input combinations, including visual, vestibular, and somatosensory, and their interaction during balance tasks.

The VR-ComBAT also assesses the participant's center of pressure, postural sway, and weight distribution, providing quantitative data for further analysis. The authors suggest that VR-ComBAT can be a promising tool for clinical assessments and interventions for balance impairments.

7.2.2 Rehabilitation is gamified by Improfit

Improfit, a Spanish startup, is using AR technology to enhance the interactivity and enjoyment of rehabilitation exercises (Forbes, 2021). The company's AR program and computer vision technology gamify rehabilitation exercises, making them more interactive and engaging for patients. Moreover, the program provides feedback to each patient by identifying the action and evaluating it in real time, enabling patients to adjust their movements for optimal outcomes.

Improfit's technology is designed to correct posture and motion while also tracking the repetitions of each exercise (Forbes, 2021). This feature ensures that patients are performing the exercises correctly and can track their progress throughout the rehabilitation process. By making physiotherapy a more enjoyable experience, Improfit aims to increase patient engagement and motivation, leading to better outcomes. With AR technology, Improfit is providing a unique and innovative solution to improve the patient experience during rehabilitation. Their approach is aimed at addressing the challenges of traditional physiotherapy and improving the effectiveness of rehabilitation programs.

7.3 Telehealth (telerehabilitation)

The COVID-19 pandemic has dramatically increased the demand for remote therapy, particularly for regular patients and nonurgent complications (Kumar, Merchant, Reynolds, & Rickards, 2021). Physiotherapy providers have had to transform their offerings into remote services, including online consultations with physiotherapists to assess a patient's physiological health status and provide remote movement therapies. This shift towards telehealth services has been vital in enabling the management of disabilities during the pandemic and beyond.

Remote physical exercises, tele-diagnostics, and tele-pharmacy are also seeing increased demand and interest from both therapists and patients (Kumar et al., 2021). These telehealth services offer convenient and accessible solutions for patients who are unable to attend in-person physiotherapy sessions or have mobility restrictions. Moreover, these services can increase the efficiency of physiotherapy treatment by allowing patients to receive care from the comfort of their own homes.

The pandemic has accelerated the adoption of telehealth services, and it is expected to continue to grow even after the pandemic subsides (Kumar et al., 2021). The availability of remote physiotherapy services has become essential for patients and physiotherapy providers, and it is likely to remain a critical component of the healthcare system in the future.

7.3.1 Online physiotherapy software by Phyt Health

Phyt, an Indian startup, has developed a platform and app to make online physiotherapy more convenient for patients (Patil, Bhagwat, & Adawade, 2021). The startup's solution focuses on at-home rehabilitation and small exercises in office settings, allowing patients to receive physiotherapy services at their own convenience. The approach of Phyt's solution is divided into four stages: pain management and protection, stabilization, strengthening, and ongoing prevention (Patil et al., 2021). These stages are designed to help patients progress through their rehabilitation process effectively. During each stage and session, artificial intelligence (AI) algorithms guide patients remotely to achieve planned results efficiently.

Phyt's online physiotherapy solution offers several benefits, including increased accessibility, convenience, and cost-effectiveness (Patil et al., 2021). Patients can receive physiotherapy services from anywhere, at any time, without having to visit a clinic. This is especially beneficial for patients who live in remote areas or have mobility restrictions. Moreover, Phyt's solution can be used for both rehabilitation and ongoing prevention, allowing patients to maintain their physical health even after the completion of their rehabilitation process. The use of AI algorithms also ensures that each patient's progress is tracked and evaluated in real time, allowing for personalized and efficient treatment.

7.3.2 E-Rehabilitation and networking by Telewecure

Telewecure, an Iranian startup, offers telerehabilitation services along with a dedicated social networking platform to its clients (Khosravi, Vanaki, & Taghizadeh, 2019). The company's remote rehabilitation solution enables patients to easily connect with physiotherapy providers via their platform, perform exercises, and receive equipment suggestions. Telewecure's solution aims to make rehabilitation a more pleasant experience for patients by offering a network that provides an online social media site for patients and professionals (Khosravi et al., 2019). This enables free communication and exchange of experiences in groups and forums that are featured in the Telewecure network. The use of telehealth services like Telewecure's solution has increased in recent years due to their convenience, accessibility, and cost-effectiveness (Khosravi et al., 2019). Patients can receive physiotherapy services from the comfort of their own homes and can connect with physiotherapy providers from anywhere in the world.

Moreover, Telewecure's social networking platform provides patients with the opportunity to share their experiences with others who are going through similar situations. This can lead to improved mental health outcomes and an increased sense of community and support for patients (Khosravi et al., 2019). Overall, Telewecure's remote rehabilitation solution is a promising development in the field of physiotherapy, offering patients a convenient and enjoyable way to receive treatment and support.

7.4 Rehabilitation wearables

In recent years, the use of health monitoring wearables and devices in rehabilitation has become increasingly prevalent. According to Soltani, Shirazi, and Abbasi (2021), wearable technology allows for more flexibility in collecting and analyzing patient data, which can later be used by rehabilitation platforms and professionals. By providing real-time data on a patient's physical activity levels and progress, wearables have become an important tool for virtual therapies and consultations. In addition, the integration of wearables with mobile technologies has led to the development of downloadable smartphone and tablet applications that can provide guidance through rehabilitation exercises and plans. These applications have the potential to increase patient engagement and motivation by providing immediate feedback and tracking progress (Soltani et al., 2021). Overall, the use of health monitoring wearables and devices in rehabilitation has significant potential for improving patient outcomes and facilitating remote rehabilitation services.

7.4.1 Smart Ms3 manufactures electromyography wearable sensors

Smart Ms3 is a US-based startup that produces wearable electromyography (EMG) sensors for tracking patient status at any given time (Smart Ms3, n.d.). EMG and musculoskeletal (MSK) sensors are used to provide precise insights into muscle activation patterns (Smart Ms3, n.d.). The sensors are wireless and are designed to track any muscle group on a patient's body (Smart Ms3, n.d.). Presently, Smart Ms3 provides applications for knee, lower back, and shoulder rehabilitation (Smart Ms3, n.d.). By providing detailed data on muscle activity, Smart Ms3 wearable sensors help physiotherapists tailor rehabilitation programs to meet the individual needs of their patients.

7.4.2 Denton creates 3D movement tracking

Denton, a UK-based startup, has developed a wearable 3D tracking technology for monitoring patients' movement during rehabilitation. The 3D sensors can be easily attached using straps on key body parts such as limbs, torso, and head. The remote connection between the sensors and the Denton app enables the monitoring of a patient's adherence levels, range of movement, exercise form, and pain intensity throughout the duration of their rehabilitation program (Denton, n.d.). This technology provides healthcare professionals with detailed data and insights, allowing them to better understand their patients' progress and tailor their treatment plans accordingly.

7.5 Rehabilitation robotics

Huang, Lv, Wei, and Zheng (2016) emphasized the role of robotic technology in the rehabilitation of patients. Robotics has become an important component in the recovery process, providing various solutions to improve rehabilitation therapy. Robotics aid in movement regeneration and accelerate progress in rehabilitation plans. Robotic exoskeletons, for example, can help patients perform daily tasks without pain and rebuild neural connections. Additionally, startups are developing lightweight wearable robots that alleviate physical disabilities and improve the quality of life for those in need of support. The integration of robotic technology with rehabilitation therapy offers a promising future for patients in the rehabilitation process (Huang et al., 2016). The field of lower limb exoskeletons has seen significant advancements in recent years. One notable development is the creation of exoskeletons that allow users to walk more naturally and with less energy expenditure. These exoskeletons use advanced algorithms and sensors to detect the user's gait and adjust the amount of assistance provided accordingly (Veneman, Kruidhof, Hekman, Ekkelenkamp, & Van der Kooij, 2007). Another recent development in lower limb exoskeletons is the use of soft robotics technology. Soft exoskeletons have several advantages over traditional rigid exoskeletons, including increased comfort and flexibility. This technology has the potential to improve the mobility of individuals with lower limb impairments and disabilities (Polygerinos, Wang, Galloway, Wood, & Walsh, 2015).

Furthermore, researchers have been working on improving the control systems for lower limb exoskeletons. Advanced control systems can enable users to perform more complex movements, such as climbing stairs or walking on uneven terrain (Van der Kooij & Grootenboer, 2011). In addition, there have been efforts to create lower limb exoskeletons that are more affordable and accessible. Startups such as Roam Robotics and Cyberdyne have developed exoskeletons that are aimed at the consumer market, allowing individuals to purchase and use these devices in their daily lives (Fergeson, 2021). Overall, these recent developments in lower limb exoskeleton technology hold great promise for improving the mobility and quality of life of individuals with lower limb impairments and disabilities.

7.5.1 Exoskeleton

Exoskeletons have emerged as an innovative technology to assist and augment human movement. They have gained immense popularity in various fields such as rehabilitation, sports, and military applications. With the advancements in technology, there have been numerous changes and improvements in the design and function of these devices. This report aims to identify and discuss the latest trends in lower limb exoskeletons.

7.5.1.1 Lightweight design
The latest trend in lower limb exoskeletons is the development of lightweight designs. Exoskeletons are becoming more comfortable and wearable for longer periods of time without causing fatigue or discomfort. According to recent research by Del Din et al. (2020), exoskeletons with a lightweight design have shown significant improvements in user mobility and physical activity levels.

7.5.1.2 Soft robotics
Soft robotics technology is being used to develop exoskeletons that are more flexible and adaptable to different body shapes and sizes. These devices are made from soft and flexible materials, which can be more comfortable to wear and reduce the risk of injury. According to Polygerinos et al. (2015), soft robotic exoskeletons have shown promising results in rehabilitation applications.

7.5.1.3 Sensor technology
Exoskeletons are increasingly being equipped with sensors that can monitor the user's movement and provide feedback to adjust the device accordingly. This helps to optimize performance and reduce the risk of injury. According to recent research by Wang, Yang, Liu, Huang, and Lian (2020), sensor-based exoskeletons have shown significant improvements in gait patterns and balance control. Multiple points were sensors that are located to acquire continuous communication feedback based on motion and load.

7.5.1.4 Neurological control
Some exoskeletons are being designed to be controlled by the user's brain waves or other neurological signals, allowing for more natural and intuitive movement. According to a study by Vuckovic et al. (2016), exoskeletons with neurological control have shown significant improvements in user mobility and quality of life.

7.5.1.5 Hybrid systems
Hybrid exoskeletons that combine mechanical and electrical systems are becoming more common. These devices can provide greater power and control than purely mechanical exoskeletons, while also being more lightweight and efficient than purely electrical systems. According to recent research by Noda et al. (2019), hybrid exoskeletons have shown significant improvements in user mobility and physical performance.

7.5.1.6 Personalization

There is a growing trend towards developing exoskeletons that are tailored to the specific needs and abilities of individual users. This can help to optimize performance and reduce the risk of injury. According to a study by Zhang et al. (2020), personalized exoskeletons have shown significant improvements in user comfort and overall performance.

In summary, the recent developments in upper and lower limb exoskeletons focus on enhancing their practicality, comfort, and effectiveness through trends such as lightweight design, soft robotics, sensor technology, neurological control, hybrid systems, and personalization. These advancements aim to extend the use of exoskeletons across various applications. With ongoing progress in technology, it is expected that more innovative and sophisticated exoskeleton designs will emerge in the future. Exoskeleton technology has evolved significantly, and researchers are continually exploring new avenues to improve the performance of upper and lower limb exoskeletons. Lightweight design is a current trend aimed at reducing the burden of heavy loads on the user while improving their maneuverability (Galle, Malcolm, & Derave, 2021). Soft robotics is another trend that involves designing exoskeletons with materials that closely resemble human tissue, providing a more comfortable and flexible interface between the user and the exoskeleton (Yap, Lim, Nasrallah, Goh, & Yeow, 2015). Sensor technology is also a crucial trend that allows for real-time monitoring and feedback of the user's movements, ensuring safe and efficient use of the exoskeletons (Bishop, Stein, & Stein, 2017). Additionally, advancements in neurological control technology have enabled the development of more responsive exoskeletons that can adapt to the user's movement intentions (Koller, Jacobs, Ferris, & Remy, 2018). Hybrid systems combining exoskeletons and functional electrical stimulation have also emerged as a promising trend in enhancing the performance of exoskeletons (Grimm, Dietrich, Dreher, & Rinderknecht, 2020). Personalization is another trend that is gaining popularity, as exoskeletons are tailored to the unique needs and capabilities of each user. By customizing the exoskeleton to fit the user's size, weight, and specific requirements, exoskeletons can optimize the user's mobility and comfort. The current trends in limb exoskeletons aim to enhance their practicality, comfort, and effectiveness. Researchers are continually exploring innovative and sophisticated designs to expand the applications of exoskeletons further. These trends demonstrate the tremendous potential of exoskeletons in improving mobility and quality of life for individuals with upper and lower limb impairments.

7.5.2 Exoskeleton hands by Nureab

Nureab, an Egyptian startup, has developed exoskeleton hands to improve the rehabilitation and movement of patients (Nureab, n.d.). The device consists of five mechanical fingers with full motion range, and provides active, passive, and resistance rehabilitation programs. The device is especially useful for treating conditions such as quadriplegia, hemiplegia, tendonitis, fractures, and injuries using resistance therapy (Nureab, n.d.). The device has highly accurate motion tracking of up to 1 degree, and is lightweight, adaptable to different hand sizes, and user-friendly (Nureab, n.d.).

7.5.3 Soft robots by Fleming MedLab

Fleming MedLab, a Chinese startup, aims to address the current challenges of high costs and difficulty in operating conventional rehabilitation robotics through the use of novel soft robots (Chen, 2020). These robots are integrated into wearable devices such as suits or clothing, making them easy to operate and allowing practitioners to focus on patients instead of managing the device. Moreover, these soft robots can be used at home, which can improve patient recovery time. Fleming MedLab is leveraging neuroplasticity in their soft robotic solution to speed up patients' active recovery (Chen, 2020). One application of this technology is for patients who have suffered strokes or heart attacks.

7.6 Personalized pre-rehab diagnostics

To establish an effective rehabilitation plan, it is necessary to diagnose the patient's current health status. The assessment of movement and neural connections is typically performed through examinations such as gait analysis or brain tests. Startups are leveraging emerging technologies, such as sensor-fitted shoe pads and AI-powered CT scanning, to enable precise real-time measuring and assessment for individual diagnostics (Zhu, Wei, & Fu, 2020). These advanced solutions provide highly personalized diagnostics, which can enhance patient recovery and outcomes.

Sensor-fitted shoe pads, for instance, offer an unobtrusive way to collect data on foot pressure, weight distribution, and gait patterns. By analyzing these data, healthcare professionals can identify gait abnormalities, which can be used to develop customized rehabilitation plans (Muro-de-la-Herran, Garcia-Zapirain, & Mendez-Zorrilla, 2014). Similarly, AI-powered CT scanning can provide detailed and accurate imaging of the brain, which can help clinicians to diagnose and monitor neurological conditions (Chen et al., 2022). In conclusion, the latest developments in sensor technology and AI-powered imaging have opened up new possibilities for highly personalized diagnostics in rehabilitation. These solutions provide accurate real-time measuring and assessment of patients' conditions, which can be used to design tailored rehabilitation plans and improve recovery outcomes.

7.6.1 Active testing for gait by LAAF

LAAF, a US-based startup, has developed a personalized gait testing and root cause analysis solution for pain (LAAF, n.d.). The company's smart shoe pads are designed to measure various parameters such as heel strike force, step clearance, pronation, cadence, and foot progression (LAAF, n.d.). Based on the data collected, LAAF's app produces personalized insights for patients to improve their performance and technique (LAAF, n.d.). This solution offers a highly personalized approach to rehabilitation, allowing for more effective treatment and better patient outcomes.

7.6.2 Multimodal imaging by Voxel AI

Rehabilitation is an important aspect of healthcare, as more and more individuals require physiotherapy after injuries, accidents, or other health issues. To develop a successful rehabilitation plan, it is crucial to perform diagnostics of the patient's current health status. Startups specializing in utilizing novel technologies, such as sensor-fitted shoe pads and AI-powered CT scanning, among others, are paving the way for personalized diagnostics, improving patient recovery and outcomes (Jung, 2021). One of these startups is the US-based LAAF, which develops smart shoe pads for personalized gait testing and root cause analysis of pain. The startup measures a variety of aspects, such as heel strike force, step clearance, pronation, cadence, and foot progression. As a result of the analysis of measured data, LAAF uses its app to produce patient insights and helps them improve their performance and technique (Jung, 2021).

In Canada, Voxel AI (Voxel AI, n.d.) assesses brain structure, function, and physiology in patients with neurological injury or disease to guide patient therapy. The startup achieves this with the use of advanced neural analysis and multimodal imaging. Voxel provides greater insight into the health of patients' brains with highly personalized, factoring in the small modalities of the brain that differentiate the outcome and required therapy. The solution enables the accurate analysis of brain injuries to make patient recovery easier and faster (Jung, 2021). These Rehabilitation Technology Trends & Startups outlined in this report only scratch the surface of trends that have been identified during the data-driven innovation and startup scouting process. Lightweight technologies, immersive tech, and novel therapies are among the emerging technologies that will transform the sector. Identifying new opportunities and emerging technologies to implement into businesses goes a long way in gaining a competitive advantage (Jung, 2021).

7.7 Photo- and electrotherapy

Light and electric therapies have been used for a long time in rehabilitation treatments to promote healing and recovery. Recent technological advancements have made these therapies more accessible and convenient for patients. Electric and light therapies aid in muscle regeneration and blood flow stimulation, among other benefits (Chen et al., 2021). Startups have developed innovative solutions that use red light and electrode-fitted suits to facilitate rehabilitation and muscle strengthening, while also reducing the time required for recovery (Rizzi, 2020).

7.7.1 Neuro20 creates suits for muscle recovery

Neuro20 is a US-based startup that focuses on developing a solution to enhance the muscle force of patients during rehabilitation. Their solution is a form-fitting Lycra suit equipped with electrodes that provide electrical muscle stimulation and gather information using biosensors. These data are then available through a remote platform (Paez, 2022). The Neuro20 suit contains 20 stimulating electrodes embedded over large muscles covering motor points, creating involuntary contractions to bring pain relief and boost muscle recovery for users (Paez, 2022). The suit is designed to enhance muscle function and strength, particularly in patients undergoing rehabilitation after injuries or surgery. The involuntary contractions created by the stimulating electrodes help to promote blood flow and reduce muscle atrophy during recovery (Paez, 2022).

The Neuro20 suit's remote platform allows clinicians and patients to monitor the progress of rehabilitation, track muscle activation, and customize the level of stimulation according to the patient's needs (Paez, 2022). This personalized approach to rehabilitation is crucial in ensuring the best possible outcomes for patients. Moreover, Neuro20 technology brings benefits to patients and physiotherapists by reducing the time required for rehabilitation and providing a less painful method of recovery (Paez, 2022). Overall, the Neuro20 suit is an innovative solution for rehabilitation that combines electrical muscle stimulation and biosensors to enhance muscle function and strength. The technology brings numerous benefits to patients and physiotherapists, including pain relief, faster recovery, and a more personalized approach to rehabilitation. The Neuro20 suit is an example of how startups are utilizing advanced technologies to transform the rehabilitation industry.

7.7.2 LUMINOUSRED Advances Red Light Therapy

In recent years, Austrian startup LUMINOUSRED has emerged as a key player in the field of rehabilitation technology by developing innovative solutions utilizing red light therapy. The company has created a specialized treatment lamp that is FDA-approved and features low-electromagnetic fields and anti-flicker technology (LUMINOSRED, n.d.). Red light therapy is a noninvasive treatment that utilizes specific wavelengths of red light to penetrate the skin and stimulate cell growth and repair (Hamblin, 2017). LUMINOUSRED's red light therapy lamps are designed to improve blood flow and support muscle regeneration, both of which are essential in the rehabilitation process

(LUMINOSRED, n.d.). The lamps are specifically designed for at-home use and are available in high intensity (100 w/cm^2) and different wavelengths (660 nm + 850 nm), allowing for customized treatment options (LUMINOSRED, n.d.).

Studies have shown the efficacy of red-light therapy in improving muscle function and reducing muscle fatigue in patients with muscle injuries (Leal Junior et al., 2010; Sawasaki et al., 2015). Furthermore, the therapy has been shown to be effective in treating chronic pain, inflammation, and skin conditions (Hamblin, 2017). In summary, LUMINOUSRED's red light therapy lamps represent a promising innovation in the field of rehabilitation technology. With their FDA-approved design, low-EMF and anti-flicker technology, and customizable treatment options, these lamps have the potential to significantly improve patient outcomes in the rehabilitation process.

7.8 Artificial intelligence

According to recent studies, the implementation of AI in rehabilitation has shown promising results for patients under the care of rehabilitation experts. The use of AI-powered solutions, particularly those based on computer vision and ML, has increased in recent years, enabling real-time insights to improve the quality of exercises and development of psychiatric plans (Kumar & Thakur, 2021).

AI-powered platforms can provide personalized and remote monitoring, as well as suggestions for further improvement based on the patient's progress. ML-powered devices, in particular, support patients throughout the rehabilitation process by analyzing the data collected from wearable sensors and other devices, providing insights into muscle activation and movement patterns, and assisting with exercises that target specific areas (Saxena & Kumar, 2021). Overall, the use of AI in rehabilitation is a promising area for future research and development, with the potential to significantly improve patient outcomes and quality of care.

7.8.1 Breathment enables AI-based remote patient management

Breathment, a German startup, has developed an AI-powered program that assists rehabilitation professionals in remotely tracking their patients' performance. This program is highly customizable and can incorporate any of the professionals' own exercises, making it remote, personalized, and data-driven for patients. By using ML algorithms, the program can analyze patients' data in real time and provide insights on their performance quality and the effectiveness of their rehabilitation plan. The platform enables remote monitoring and offers personalized suggestions for further improvement. This technology makes rehabilitation more accessible and efficient by providing remote care and improving the quality of care provided by rehabilitation professionals.

Using only a mobile device with a camera, Breathment's AI-Trainer allows your patients to perform breathing and physical exercises without your presence. An AI-Trainer guides patients through their exercises while monitoring them at the same time. It provides instant feedback on whether the patient is performing the exercise correctly by using AI-powered motion and breathing pattern tracking.

7.8.2 AI-driven rehabilitation solutions by Rootally

Rootally, a Singaporean startup, has developed an AI-powered home rehabilitation solution, AllyCare, that makes therapy more convenient and affordable (Rootally, n.d.). AllyCare utilizes mobile devices to show an AI-generated model of any exercise while simultaneously tracking the patient's movements, providing a user-friendly solution for rehabilitation. By analyzing the rehabilitation sessions and tracking the patient's progress, AllyCare provides insights into patients' progression, allowing therapists to customize therapy plans for better outcomes (Rootally, n.d.). Unlike other solutions, AllyCare does not require any sensors attached to the patient's body, making it more accessible for patients who may not have the equipment or ability to use more complex solutions (Rootally, n.d.).

7.8.3 ChatGPT and Bing AI

ChatGPT and Bing AI are two advanced technologies that have the potential to greatly impact the health industry. ChatGPT is an AI language model trained by OpenAI, while Bing AI is a suite of AI-powered tools and services developed by Microsoft. These technologies can be used to improve patient outcomes, optimize clinical workflows, and enhance the overall quality of healthcare services. One of the keyways in which ChatGPT and Bing AI can benefit the health industry is by improving patient communication and engagement. ChatGPT can be used to develop chatbots that can interact with patients and provide them with relevant information about their health conditions, medications, and treatment plans (Halamka & Mandl, 2020). Bing AI, on the other hand, can be used to develop voice assistants that can provide patients with personalized health recommendations and advice based on their medical history and preferences (Microsoft, n.d.).

Another important application of ChatGPT and Bing AI in healthcare is in the field of medical imaging. ChatGPT can be used to develop algorithms that can analyze medical images and identify patterns and anomalies that may be missed by human radiologists (Johnson et al., 2020). Bing AI can be used to develop tools that can help radiologists and other healthcare professionals to interpret medical images more accurately and efficiently (Microsoft, n.d.). Finally, ChatGPT and Bing AI can be used to improve clinical decision-making and optimize healthcare workflows. ChatGPT can be used to develop predictive models that can help healthcare providers to identify patients who are at high risk of developing certain health conditions and develop personalized treatment plans for them (Halamka & Mandl, 2020). Bing AI can be used to develop tools that can automate routine administrative tasks, such as scheduling appointments and managing patient records, allowing healthcare providers to focus more on patient care (Microsoft, n.d.).

In conclusion, ChatGPT and Bing AI have the potential to greatly improve the quality and efficiency of healthcare services. By enhancing patient communication and engagement, improving medical imaging analysis, and optimizing clinical workflows, these technologies can help to deliver better patient outcomes and promote the overall health and well-being of individuals and communities.

7.9 Neurofeedback

Neurofeedback, a type of brain-computer interface technology, has emerged as an innovative trend in the rehabilitation industry. This technology is based on the principle that the human brain generates electrical signals, which can be captured and analyzed in real time using wearable sensors and electrostimulation devices. The use of neurofeedback aims to facilitate noninvasive rehabilitation for patients following stroke, concussion, or in the management of pain (Müller, Mochty, & Heinrich, 2019). Neurofeedback has shown promise in the rehabilitation of patients with neurological disorders such as Parkinson's disease, epilepsy, and traumatic brain injury (TBIs) (Ros, Baars-Elsinga, Metzemaekers, & Münte, 2019). For instance, studies have demonstrated that neurofeedback training can lead to significant improvements in cognitive and motor functions in patients with Parkinson's disease (Kleih, Nijboer, Halder, & Kübler, 2015). Additionally, neurofeedback has been used to improve memory and attention in patients with epilepsy and TBIs (Ros et al., 2019).

Furthermore, the potential of neurofeedback extends beyond rehabilitation to the enhancement of brain and neural performance in individuals with long-term neurological disorders. For example, neurofeedback has been shown to improve cognitive function and attention in healthy individuals, suggesting potential applications for performance enhancement in healthy populations (van Boxtel & Gruzelier, 2014). Neurofeedback has emerged as a promising trend in the rehabilitation industry, offering noninvasive solutions for patients with neurological disorders. With further research, this technology may also have the potential to enhance cognitive and neural performance in healthy individuals.

7.9.1 Remote neurofeedback solutions by Divergence Neuro

Divergence Neuro, a Canadian startup, has developed an innovative solution that brings neurofeedback to online therapy. The solution includes a wearable headset device that provides real-time neurofeedback of patients' brain functions and pain levels. The device is designed to fit any user and works in conjunction with the startup's platform and mobile app (Divergence Neuro, 2022). The Divergence Neuro platform is specifically designed to assist therapists in designing treatment sessions that incorporate a variety of quantitative and qualitative assessments in line with predefined or custom treatment protocols. These protocols may include neurofeedback training or neuromeditation, depending on the patient's needs (Divergence Neuro, 2022).

The mobile app offered by Divergence Neuro allows patients to access the treatment programs assigned to them by professionals. Patients are also guided through the neurofeedback or neuro-meditation protocols that their therapists have assigned them to complete. The app enables patients to track their progress and communicate with their therapist in real time, providing a more personalized and collaborative approach to online therapy (Divergence Neuro, 2022). The integration of neurofeedback into online therapy represents an innovative approach to providing noninvasive and personalized treatment to patients. The use of wearable technology to provide real-time feedback allows therapists to monitor and adjust treatment programs as needed, improving patient outcomes. Moreover, the availability of a mobile app that guides patients through their treatment programs and enables them to communicate with their therapist in real time increases treatment adherence and engagement (Divergence Neuro, 2022). In conclusion, the Divergence Neuro solution represents a promising innovation in the field of online therapy, providing an integrated and personalized approach to treatment. With further research, this technology has the potential to transform the way we approach the treatment of mental health conditions.

7.9.2 Neurostimulation headset by Exsurgo

Exsurgo, a New Zealand-based startup, has developed a novel wearable device for pain and rehabilitation management. The device, called AXON, tracks brain activity data using electroencephalography (EEG) and AI, and displays this information on the user's mobile device. By providing real-time visual feedback, the user can learn how to modulate their pain by recognizing and modifying their response, resulting in a significant decrease in pain levels (Exsurgo, 2021). The AXON device represents a promising innovation in the field of pain management and rehabilitation. By using EEG technology, it provides a noninvasive and real-time assessment of the patient's brain activity, enabling clinicians to tailor treatment programs to meet individual patient needs (Lugo, Nguyen, & de Castro Campos, 2021). The use of AI to analyze this data further enhances the device's ability to provide personalized treatment, as it can identify patterns and trends in the patient's brain activity that may indicate specific treatment needs (Lugo et al., 2021).

The device's real-time visual feedback and the ability to track progress over time also promotes patient engagement and adherence to treatment protocols. Patients are empowered to take control of their own pain management and rehabilitation, as they can proceed with their treatment from the comfort of their homes, without the need for frequent visits to healthcare providers (Exsurgo, 2021). This increased patient autonomy is likely to result in improved treatment outcomes and a reduction in healthcare costs associated with pain management and rehabilitation. In conclusion, the AXON wearable device represents an innovative and promising approach to pain management and rehabilitation. Its use of EEG and AI technologies, coupled with real-time visual feedback, provides clinicians with the ability to tailor treatment programs to meet individual patient needs. Moreover, its convenience and noninvasiveness enable patients to take control of their own treatment, promoting engagement and adherence to treatment protocols.

7.10 Technology for lightening/unweighting

During the convalescence period, therapy requires a gradual introduction of body weight to the limbs during exercise to become effective. To unweight the body, several technologies are being used, including water, vacuum, and supporting devices that redirect forces to nondamaged body parts (Li, Rong, & Xu, 2020). These technologies can be particularly useful for patients who have suffered injuries or undergone surgeries that limit their ability to bear weight on their limbs. One technology that has been developed by startups to support rehabilitation is the vacuum treadmill. This device uses negative pressure to unweight the patient's lower limbs and allow for controlled movement and exercise (Alekseev, Chepik, & Leonova, 2020). By reducing the pressure on the anterior limbs, the vacuum treadmill allows patients to engage in rehabilitation exercises without bearing their full body weight, which can be particularly beneficial for patients with lower limb injuries.

In addition to vacuum treadmills, supportive devices such as walkers can also be used during rehabilitation to redirect forces to nondamaged body parts. Walkers can provide a stable support for patients during exercise, allowing them to gradually increase weight bearing on their injured limbs (Shen et al., 2021). This can be particularly useful for patients who are recovering from lower limb injuries or surgeries. Overall, the use of technologies such as vacuum treadmills and walkers during rehabilitation can be beneficial for patients during the convalescence period. These devices allow for a gradual introduction of body weight to the limbs during exercise, promoting effective rehabilitation and preventing further injuries. Startups are driving innovation in this field, developing new and improved technologies that can support patients during their recovery process.

7.10.1 Lightweight passive exoskeleton built by MEBSTER

MEBSTER, a Czechia-based startup, has developed UNILEXA, a nonrobotic exoskeleton equipped with multiple sensors to aid in lower limb unweighting during rehabilitation. UNILEXA helps patients reduce the stress on their lower limbs and helps them move more easily. The startup aims to create affordable and useful home assistive devices that enable people to walk again. UNILEXA is already being utilized in healthcare centers to track patient conditions and rehabilitation progress (MEBSTER, n.d.). Lower limb unweighting is a crucial part of rehabilitation, allowing for a gradual introduction of body weight to the limbs during exercise to become effective (Lohse, Lang, & Boyd, 2014). A variety of technologies are used to unweight the body, including water, vacuum, and supportive devices, that redirect forces to nondamaged body parts (Janssen, Bussmann, & Stam, 2010). By developing UNILEXA, MEBSTER contributes to the trend of developing innovative solutions that help patients recover and regain mobility during the convalescence period.

7.10.2 Microgravity treadmills developed by Boost Treadmills

Boost Treadmills, a US-based startup, is providing an innovative solution to support lower limb recovery by developing a vacuum-powered unweighting treadmill. The company's first model, boost 1, provides adjustable levels of force that patients must bear on their legs and lower back during rehabilitation, significantly impacting rehabilitation time and pain endurance. The solution relies on creating a vacuum chamber to reduce gravitational force, making physiotherapy more comfortable and effective. As the patient moves, the system uses the vacuum to unweight the body, reducing the impact on the lower limbs while still allowing natural movement. Boost Treadmills is now developing the Boost 2, which includes new features, such as simplified controls, gait analysis, and pre-programmed workout management, to further improve the user experience. Using technology such as vacuum unweighting treadmills has shown to be effective in supporting rehabilitation by reducing the pressure on the lower limbs during exercise. This technology has been utilized by other startups such as MEBSTER, which developed the UNILEXA, a nonrobotic exoskeleton that helps patients reduce stress on their lower limbs, making movement easier and more comfortable (MEBSTER, n.d.).

7.11 Analytics and big data

Medical and physiotherapy data have the potential to play a critical role in shaping decisions that impact a patient's lifetime. These data can be leveraged to inform experts of patient progress through various procedures and conditions over time. With the help of ML algorithms, big data extracted from medical and physiotherapy can also drive precision medicine and the sale of therapy-assistive devices such as exoskeletons and crutches. Startups operating in this field are tackling the challenge of managing big data by offering innovative health platforms that make data more accessible and secure. In recent years, big data analytics has become an essential tool in healthcare research, providing insights that can improve patient outcomes and reduce healthcare costs. According to a report by Grand View Research, the global healthcare big data analytics market is expected to grow at a compound annual growth rate of 22.7% from 2020 to 2027 (Grand View Research, 2020). This growth is driven by the increasing adoption of electronic health records, the proliferation of digital health devices, and the need to improve patient care.

Startups are taking advantage of this growth opportunity by providing innovative health platforms that use big data to provide insights and recommendations. One such startup is nference, which uses AI to extract insights from biomedical literature, clinical trials, and patient data. This platform enables healthcare professionals to make data-driven decisions and improve patient outcomes (nference, n.d.). Another startup, Medopad, provides a digital health platform that collects patient data from wearables, medical devices, and electronic health records. These data are analyzed using ML algorithms to provide insights on patient progress and treatment efficacy. Medopad's platform is also used to develop personalized treatment plans for patients, improving their chances of a successful recovery (Medopad, n.d.). In conclusion, startups in the healthcare industry are leveraging big data and ML algorithms to provide innovative health platforms that improve patient outcomes and reduce healthcare costs. These platforms offer a way to manage and analyze medical and physiotherapy data effectively while ensuring data security and privacy. As the demand for precision medicine and therapy-assistive devices increases, startups operating in this field are well positioned to play a crucial role in shaping the future of healthcare.

7.11.1 Interdisciplinary dataset developed by Precise4Q

Precise4Q, a German startup, aims to combat stroke by utilizing a multidimensional data-driven predictive simulation model to enable personalized stroke treatment. The startup aims to address patients' needs across four stages: prevention, acute treatment, rehabilitation, and reintegration (Bisdas, Ahmadipour, & Spyridopoulos, 2021). By utilizing a variety of multidisciplinary sources such as genomics, microbiomics, biochemistry, and imaging, Precise4Q can create a robust dataset that allows for personalized stroke treatment (Bisdas et al., 2021). The utilization of such data-driven approaches in healthcare has the potential to lead to significant reductions in healthcare costs, both direct and indirect, by enabling the development of personalized prevention, treatment, and rehabilitation programs (Chawla & Kuipers, 2020). Precise4Q approach is an example of the benefits of utilizing big data to improve patient outcomes in healthcare.

7.11.2 Data-as-a-service for rehabilitation

Forhealth, an Indian startup, aims to collect data from various sources to create a multidimensional dataset for rehabilitation companies and patients. The company's software as a service (SaaS) platform gathers data from hospitals, gyms, and old age homes, providing feedback to these institutions. By utilizing ML algorithms, Forhealth seeks to develop predictive models that will enhance treatment recommendations for patients undergoing rehabilitation. Data play a significant role in improving the effectiveness of rehabilitation treatment, and startups like Forhealth aim to address the challenges of data management in the healthcare industry. According to a report by the World Health Organization (WHO), collecting and analyzing data can lead to better healthcare outcomes (World Health Organization, 2019). By leveraging various sources of data, Forhealth aims to provide rehabilitation companies and patients with valuable insights that will improve treatment outcomes.

The use of ML algorithms in healthcare has gained significant attention in recent years. A study by Rajkomar et al. (2018) demonstrated the potential of ML algorithms in predicting clinical outcomes in patients. By analyzing a large amount of data, the authors were able to develop models that predicted outcomes such as mortality, readmission rates, and length of stay. This study highlights the potential of ML algorithms in healthcare and how they can aid in making more informed decisions about patient care. Startups like Forhealth are also looking to provide personalized rehabilitation programs to patients. According to a report by the National Institute of Neurological Disorders and Stroke (National Institute of Neurological Disorders and Stroke, 2019), personalized rehabilitation programs can help improve patient outcomes. By utilizing data from various sources, Forhealth aims to create personalized treatment plans that will help patients recover more effectively.

7.11.3 Impact of rehabilitation technology in 2023

Rehabilitation technologies have made significant strides in recent years, and their impact on healthcare is likely to increase in 2023. One of the key benefits of rehabilitation technologies is their ability to provide remote care and monitoring, enabling patients to receive personalized care from the comfort of their own homes (Bower, 2020). This is particularly important considering the COVID-19 pandemic, which has made it difficult for patients to access healthcare facilities. Another major impact of rehabilitation technologies is their ability to provide real-time data and insights on patients' progress, allowing for personalized treatment plans and better outcomes (Gutenbrunner, Ward, Chamberlain, Kiekens, & Imamura, 2018). This is made possible with AI and ML, which can analyze large amounts of data and provide recommendations for further improvement (Zhang et al., 2020).

In addition, startups such as Neuro20, LUMINOSRED, and Breathment are developing innovative solutions for rehabilitation using electrical and light therapies, as well as AI-powered platforms for remote monitoring and personalized care (Paez, 2022; Breathment, n.d.; Breathment, n.d.; Luminosred, n.d.). These solutions have the potential to streamline the rehabilitation process, reduce costs, and improve patient outcomes. However, there are also challenges associated with the use of rehabilitation technologies, such as data privacy concerns, lack of access to technology in underserved communities, and the need for healthcare professionals to be trained in their use (Gutenbrunner et al., 2018). Addressing these challenges will be essential to fully realize the potential of rehabilitation technologies in 2023 and beyond.

7.11.4 Limitations of rehabilitation technology in 2023

Rehabilitation technologies have made significant strides in recent years, offering a range of benefits such as remote monitoring, personalized therapy plans, and real-time data analysis. However, despite their potential advantages, these technologies also have certain limitations that need to be addressed. One major limitation is the lack of access to rehabilitation technologies in low-income areas and developing countries. The cost of these technologies can be prohibitive, making them inaccessible to people who may need them the most (Dizon et al., 2021). Additionally, the use of these technologies requires access to reliable internet connectivity and the necessary devices, which may not be available in remote or underserved areas (Dizon et al., 2021). Another limitation is the lack of regulatory oversight and standardized protocols for the use of these technologies. The fast-paced development and deployment of these technologies have outpaced regulatory agencies' ability to keep up, leading to potential safety concerns and inadequate testing (Alhussein et al., 2021).

Additionally, the use of rehabilitation technologies requires patients to have a certain level of technical proficiency and comfort with using devices and software, which may not be the case for all patients (Zhang et al., 2021). This can lead to decreased adherence and effectiveness of rehabilitation programs, as patients may struggle with using the technologies correctly. Finally, the data collected from rehabilitation technologies may be subject to privacy and security risks, particularly when stored on cloud-based platforms. This raises concerns about the potential misuse and unauthorized access to sensitive patient information (Dizon et al., 2021). In conclusion, while rehabilitation technologies offer significant benefits, including remote monitoring, personalized therapy plans, and real-time data analysis, they also face certain limitations that need to be addressed. These limitations include lack of access in low-income areas, lack of regulatory oversight, technical proficiency requirements, and privacy concerns.

7.12 Conclusion

In recent years, the healthcare industry has witnessed a growing interest in the use of digital technology for lower limb injuries rehabilitation. This technological advancement has the potential to revolutionize how patients receive care and recover from injuries. One significant trend in this area is the use of wearable technology, such as smart sensors and activity trackers. These devices are increasingly being used to monitor and track patients' progress during rehabilitation, providing real-time feedback to both patients and healthcare providers. By tracking patients' movements and activities, wearable technology can help healthcare providers make better decisions about treatment and rehabilitation. Another trend that is gaining momentum is the use of VR technology in rehabilitation. By creating immersive and engaging experiences that simulate real-life scenarios, VR can challenge patients and help them to improve their motor skills and balance. For example, patients with lower limb injuries can use VR technology to simulate activities such as walking or climbing stairs, helping them to regain strength and mobility.

Telehealth has also emerged as a popular option for lower limb injuries rehabilitation. This technology allows patients to receive care remotely from their homes, reducing the need for in-person visits to healthcare facilities. Telehealth can include video consultations with healthcare providers, virtual rehabilitation sessions, and remote monitoring of patients' progress. This not only provides convenience to patients but also ensures that they receive the care they need without the need for travel. Robotics is another area where digital technology is being increasingly used for lower limb injuries rehabilitation. Robotic devices can aid and support to patients during exercises and activities, helping them to regain strength and mobility while minimizing the risk of further injury. These devices can also track patients' progress and provide feedback to healthcare providers. Finally, AI is being used to develop personalized rehabilitation programs for patients based on their specific needs and abilities. AI algorithms can analyze data from wearable devices and other sensors to provide insights into patients' progress and suggest adjustments to their rehabilitation plans. This can help healthcare providers to develop more effective rehabilitation plans that are tailored to each patient's needs.

In conclusion, digital technology applications have the potential to transform how lower limb injuries rehabilitation is conducted. The use of wearable technology, VR, telehealth, robotics, and AI provides patients with new opportunities for personalized and engaging care. As technology continues to advance and healthcare providers explore new ways to improve patient outcomes, it is likely that these trends will continue to evolve and expand in the coming years. I would like to provide special credit to StartUs insights for providing a direction and valuable data on the latest technology surrounding injury rehabilitation (Insights, 2023).

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